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Topic of Project:

Logic-Based Vending Machine Controller

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Table of Contents

- Abstract
- Introduction
- Project Specifications (First main point)
- Logic Design and Implementation (Second main point)
- Simulation Results and Analysis (Third main point)
- Conclusion
- Team Roles
- Hardware
- References

Abstract

This project presents the comprehensive design and implementation of a digital logic-based controller for a simplified vending machine. The system is engineered to handle transactions for a product priced at 15 currency units by accepting two coin denominations: 5 and 10. Using combinational logic circuits, the controller evaluates three binary inputs in real-time to determine when the threshold for dispensing is met. Furthermore, the system is designed to provide change when an overpayment occurs, specifically returning 5 units when a total of 20 is inserted. The design was validated through simulation in Logisim, confirming the accuracy of the Boolean logic and ensuring the reliability of the dispensing and change-return mechanisms.

Introduction

Vending machines are ubiquitous examples of automated decision-making systems. At their core, they rely on precise logic to ensure that products are only dispensed upon the receipt of correct payment and that excess funds are accurately returned to the customer. This project focuses on the development of a logic-based controller that simulates these fundamental functions using digital gates.

The primary function of the system is to monitor three payment inputs: two 5-unit coin slots and one 10-unit coin slot. The controller must logically combine these signals to activate a 'Dispense' output when the total sum reaches 15. Additionally, it must handle the 'Change' case where a user might insert two 5-unit coins and one 10-unit coin, totaling 20. The importance of this project lies in applying theoretical concepts of Boolean algebra and Karnaugh Maps to a practical, real-world scenario, demonstrating how complex behaviors can be reduced to simplified digital logic circuits.

Project Specifications (First main point)

The controller is defined by its interactions with external components, categorized into binary inputs and digital outputs.

System Inputs

- Input A: A binary signal representing the insertion of a 5-unit coin.
- Input B: A binary signal representing the insertion of a second 5-unit coin.
- Input C: A binary signal representing the insertion of a 10-unit coin.

System Outputs

- Out Signal: A logical output that triggers the product dispensing mechanism (LED indicator). It activates when the sum of active inputs equals or exceeds 15.
- Change Signal: A logical output that activates the coin return mechanism (7-segment display). It is specifically triggered when the total input value is 20, signifying a 5-unit refund is required.

Logic Design and Implementation (Second main point)

The design process followed a standard digital logic workflow, starting from a Truth Table and proceeding to minimized Boolean equations.

Truth Table

A (5 coin)	B (5 coin)	C (10 coin)	Out (Dispense)	Change (Return 5coin)
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	**1**	0
1	0	0	0	0
1	0	1	**1**	0
1	1	0	0	0
1	1	1	**1**	**1**

Minimized Boolean Equations

$$Out = C.(A + B)$$

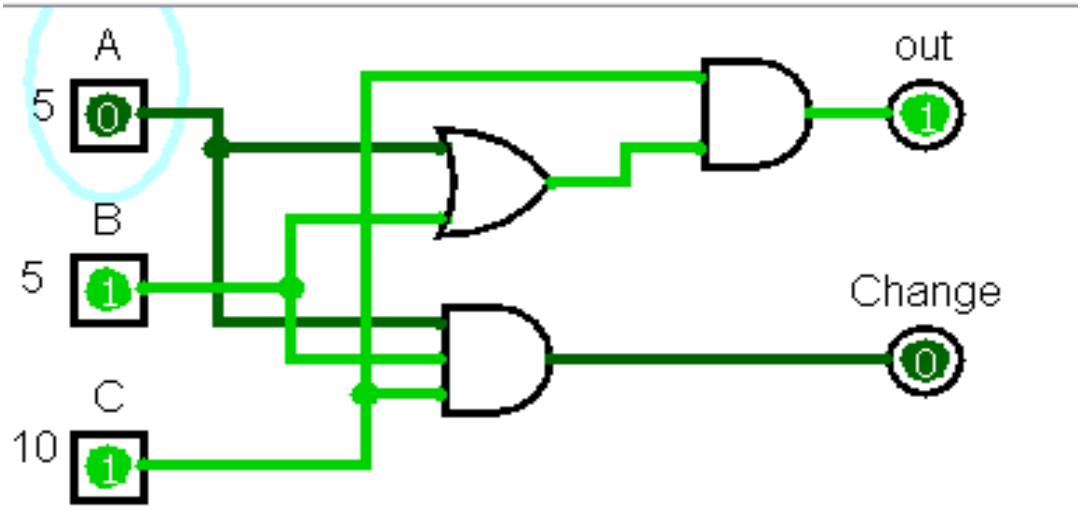
$$Change = A \cdot B \cdot C$$

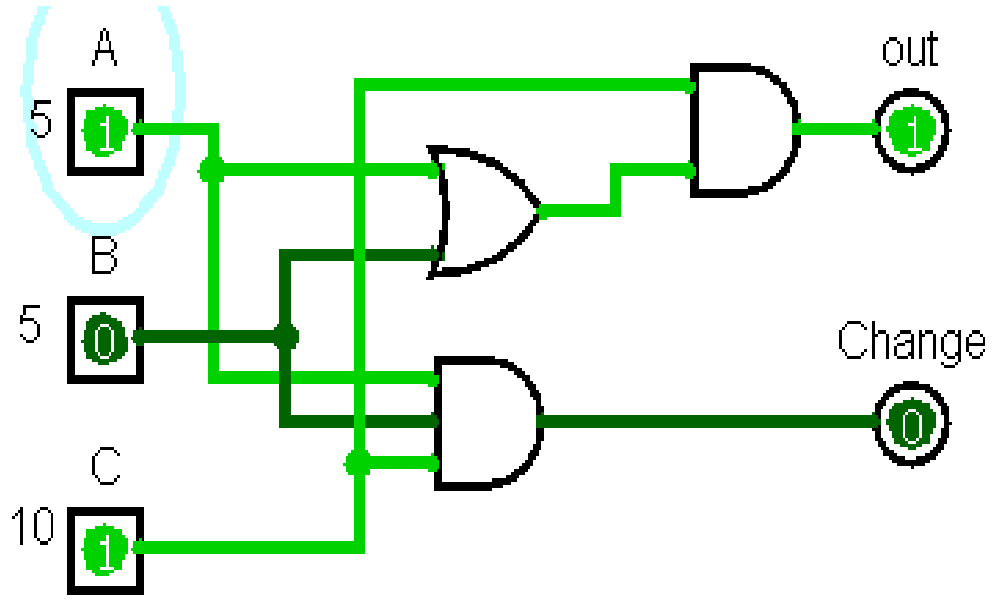
Simulation Results and Analysis (Third main point)

The logic was implemented in Logisim using AND and OR gates. Simulation was conducted for all key payment scenarios.

Scenario 1: Exact Payment (Total = 15)

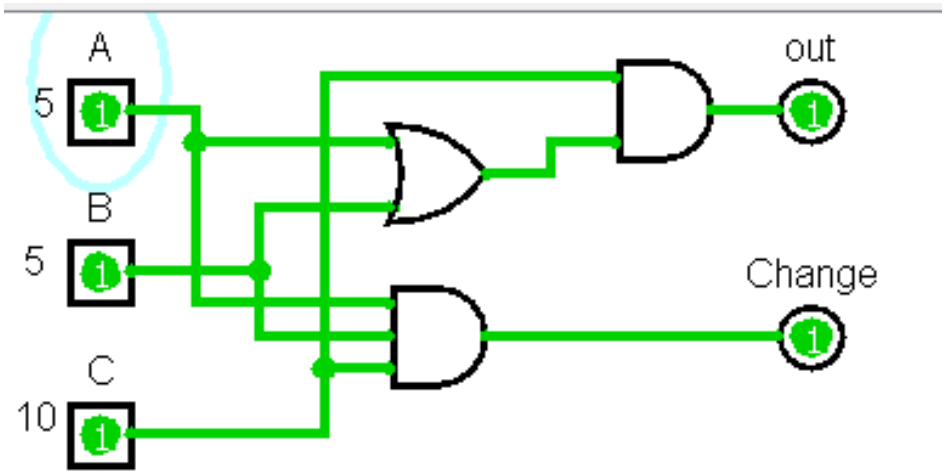
As shown in the simulation screenshots, when Input A (5) and Input C (10) are set to high, the 'Out' LED activates while 'Change' remains inactive.





Scenario 2: Overpayment (Total = 20)

When all inputs (A, B, and C) are set to high, the simulation shows both the 'Out' and 'Change' signals activating.



Conclusion

The logic-based vending machine controller project successfully demonstrated the application of combinational logic to solve a practical automation problem. The simulation results consistently aligned with the theoretical design, proving the robustness of the logic.

Team Roles

Issa – circuit drawing & hardware

Ahmed – Simulation & testing

Abdalla – Report writing & truth table

* Hardware

List of components	
Breadboard	4
IC 7432 (OR GATE)	1
IC 7408 (AND GATE) 2 input	1
IC 7411 (AND GATE) 3 input	1
IC 7404 (NOT GATE)	1
Press 4Pin Tactile Switches	3
Resistors	8
7 Segment Common Anode	1
LED	1
Jumper wires	

Explanation of how the hardware works

The hardware circuit is wired to directly reflect our Boolean equations. We used 4-pin tactile switches as our coin inputs (A, B, and C), providing a 5V signal (Logic 1) when pressed.

For the 'Out' signal (Dispensing): First, we routed inputs A and B into the 7432 IC (OR gate). Then, the output of this OR gate is routed along with input C into the 7408 IC (AND gate). The final output from this AND gate goes to an LED to visually indicate that the product is dispensed.

For the 'Change' signal: We routed inputs A, B, and C through the 7408 IC (AND gate) logic. The final output of this operation is connected to the 7-segment display to show the number '5', indicating the returned change.

Differences between the simulation and real circuit

While the Logisim simulation represents ideal conditions, the real hardware implementation required physical wiring and power management. We had to provide a 5V power supply (VCC and GND) to the 7432 (OR) and 7408 (AND) ICs, which isn't needed in simulation. Additionally, using physical tactile switches required stable wiring to avoid floating inputs, and we had to manually map the specific pins of the Common Anode 7-segment display to correctly show the number '5' for the change signal.

References

[1] Morris Mano, M. and Ciletti, M. D. (2017). Digital Design: With an Introduction to the Verilog HDL, VHDL, and SystemVerilog. 6th Edition. Pearson.

[2] Logisim Simulation Documentation. (2026). Digital Circuit Design and Simulation Software.

[3] Faculty Course Materials. (2026). CSC122: Logic Design Course Notes. Future University in Egypt.